CampusFlow 3D

A Web-Based University Transit Digital Twin Platform

Software Engineering and Information System Design Lab

Course Code: CSE-3104

Submitted by:

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Team Members and Roles

Name	Student ID	Role & Responsibilities
Mohammod Abdullah Azrof Sadim	230102026	**Team Leader & Systems Architect** - Project coordination, agile management, system architecture design, Git collaboration, and ensuring decoupled frontend/backend communication.

Araf M N Ishtiaq	230102027	**Backend Engineer** - Python (FastAPI) backend development, API routing, simulation loop creation, and managing the core transit logic.
Yeasin Arafat	230102022	**AI & Machine Learning Engineer** - Scikit-Learn model training, mathematical logic for "Self-Sharpening" traffic ETAs, and determining 100/200% capacity thresholds.
Sajal Tikadar	230102031	**Frontend & Spatial UI Developer** - Interface design, React.js implementation, Mapbox GL JS integration, and rendering interactive 3D bus models on the map.
Patha Sarathi Biswas	230102013	**Database & Real-Time Systems** - PostgreSQL database schema design, historical travel data logging, and managing Socket.IO WebSockets for zero-latency telemetry streaming.

Sojib Ahmed	230102008	**QA Tester & Deployment Engineer** - Writing automated Geofencing logic, bug tracking, system stress-testing, and deploying the application to cloud platforms (Vercel/Render).

Project Idea

Title

CampusFlow 3D - A Web-Based University Transit Digital Twin Platform

Overview

CampusFlow 3D is an innovative, software-based Digital Twin engineered to digitize and centralize the logistics of the University of Barishal's transit network. It is specially designed for university students and the transport pool administration.

Through CampusFlow 3D, students can visually track the university's 20-bus fleet across six major routes via a live, 2.5D interactive WebGL spatial environment. The system focuses on predictive intelligence, offering historically sharpened ETAs, dynamic route availability, and AI-driven capacity tracking. Transport administrators maintain complete oversight of the fleet, schedules, and historical analytics from a centralized dashboard. It transforms the traditional transit workflow—previously dependent on static paper schedules and guesswork—into a modern, proactive, and data-driven digital ecosystem.

Problem Statement

In universities across Bangladesh, transit management processes remain completely static and largely blind. At the University of Barishal, buses operate via designated stops without live tracking. Students are forced to guess arrival times based entirely on static, paper-based departure schedules, causing severe spatial blindness.

This disconnect creates the "Blind Waiting" phenomenon. Students regularly wait 30 to 40 minutes at major traffic bottlenecks (like Nothullabad or Amtola More) without knowing the capacity of the approaching vehicle. Discovering an arriving bus is at 100% capacity with no standing room left results in severely wasted time, frequently causing students to miss urgent classes and exams. Furthermore, the system suffers from "Ghost Bus" uncertainty; if a peripheral route is inactive on an off-day or breaks down, students wait in vain. Administrators also lack empirical data on when and where buses become overloaded, making efficient fleet optimization mathematically impossible.

Motivation

The motivation behind CampusFlow 3D comes from the daily, observable struggles of Barishal University students losing valuable time to unpredictable transit. We want to empower students to make executive, data-driven decisions. If a student sees an approaching bus glowing "Red" (overloaded), they can instantly arrange local transport instead of waiting blindly.

Furthermore, this project acts as an academic incubator. While the initial focus is the university campus, the ultimate long-term vision is to scale this AI-driven Digital Twin to commercial *poribohon* bus services across Bangladesh—revolutionizing inter-city travel, eliminating blind waits at ticket counters, and establishing a unified, predictive smart-transit grid.

Feature and Design Direction

CampusFlow 3D will follow a modern, highly visual, and interactive design approach.

Design Direction

- High-contrast 2.5D Mapbox interface showcasing the Barishal, Jhalokathi, and Patuakhali routes.
- Interactive 3D .gltf bus models that expand into detail panels upon clicking.
- Color-coded visual indicators embedded directly into the map for immediate capacity recognition.
- Clean, responsive dashboard UI suitable for both mobile tracking (students) and desktop monitoring (admins).

Feature Direction

- Real-time spatial tracking of the entire active fleet.
- AI-driven Capacity Color-Coding (Yellow for 100% seating, Red for 200% cultural standing room limit).
- Self-Sharpening Machine Learning ETAs adjusting for rush-hour traffic.
- Automated Geofencing to toggle buses "In Service" or "Out of Service" automatically.

SWOT Analysis

Strengths

- Deep team understanding of decoupled architectures, WebSockets, and AI integration.
- The team directly understands the real cultural challenges (like 200% standing capacity)

in local transit.
- Highly innovative fusion of 3D spatial computing and machine learning.

Weaknesses

- Lack of physical IoT hardware and enterprise funding for a full physical prototype.
- Strict 1-month development time due to academic coursework constraints.

Opportunities

- Fills a documented academic gap in Human-Computer Interaction (HCI) for Digital Twins.
- High educational impact by saving thousands of collective student commute hours.
- Potential to scale the MVP architecture to national commercial transit fleets.

Threats

- Unpredictable third-party API limits (Mapbox free tier).
- Potential network latency affecting real-time WebSocket data streaming.

Expected Features / Requirements

Student / Passenger Profile

- Browse the live 2.5D interactive map.
- Click 3D bus models to view Route Designation, Number, and Next Stop.
- View "Self-Sharpening" ETA predictions adjusted for traffic.
- Identify overloaded buses instantly via the Yellow (100%) and Red (200%) color-coding.
- View the persistent "List of Buses" to see which routes have a Green Circle (Active) or Red Circle (Off-Duty).

Organizer / Admin Profile

- Access the centralized Transport Pool Dashboard.
- Monitor the entire active 20-bus fleet simultaneously on a global map.
- Receive high-priority automated SOS alerts with locked GPS coordinates if a bus breaks down.
- View historical Analytics Dashboards to see average crowd density per hour.
- Generate fleet impact reports to justify budget and schedule adjustments.

Special Features

- **Geofenced State Automation:** Virtual boundaries around the depot automatically toggle bus statuses, eliminating human entry error.
- **Culturally Calibrated AI:** Capacity logic respects the reality of Bangladeshi transit, only marking a bus "Red" when standing room is also exhausted.

- **Decoupled Telemetry:** Simulation loops run independently on the backend, streaming coordinates via WebSockets to keep the frontend lightweight.

Development Approach

Model: Agile

1. Planning & Requirement Analysis
2. Design (UI/UX & System Architecture)
3. Implementation / Development
4. Testing & Quality Assurance
5. Deployment
6. Maintenance & Feedback

The Agile model is selected to support iterative prototyping. Sprints will focus on distinct modules: Map rendering, Backend simulation, WebSockets, AI integration, and Analytics.

Tools and Technologies

Category	**Recommended Tools**	**Purpose / Justification**
Frontend	React.js, Mapbox GL JS, Tailwind CSS	High-performance component UI, hardware-accelerated 2.5D map rendering, and responsive styling.
Backend	Python (FastAPI)	High-speed API routing, simulation loop processing, and asynchronous task management.
Database	PostgreSQL	Relational storage for historical travel times, schedules, and

		analytics logging.
AI / ML	Scikit-Learn	Training models for time-series ETA predictions and capacity thresholds.
Real-Time	Socket.IO (WebSockets)	Zero-latency, bi-directional event streaming for live bus coordinates.
Version Control	Git + GitHub	Team collaboration, branch management, and code review.
Deployment	Vercel (Frontend), Render/PythonAnywhere (Backend)	Free-tier, continuous integration cloud hosting for the MVP.

Budget Estimation

Because CampusFlow 3D represents an enterprise-grade concept, we contrast the theoretical physical rollout costs against the actual costs required to build our software MVP.

Theoretical Enterprise Rollout Costs (20-Bus Fleet)

Category	Description	Estimated Cost (BDT)	Notes

Hardware	4G Trackers, Passenger Counting Cameras, Edge AI Compute Nodes	6,40,000	One-Time
Software	Mapbox Commercial Licenses, AWS Cloud Hosting, DB	4,00,000	Annual
Labor	Engineering Team Salaries, Installation Technicians	3,60,000	One-Time
Maintenance	Spare parts, SIM data, contingency buffer	1,00,000	Annual
Total	**Estimated Enterprise Budget**	**15,00,000 BDT**	

Actual MVP Simulation Costs

Category	**Description**	**Estimated Cost (BDT)**	**Notes**

Development	In-house coding by the 6-member student team	0	Academic
Tools	Mapbox Developer Tier, Open-Source Python, Free 3D Assets	0	Free Tiers
Hosting	Vercel (React), Render/PythonAnywhere (FastAPI)	0	Free Tiers
Total	**Estimated MVP Budget**	**0 BDT**	

Timeline

Duration	**Tasks**
Days 1-10	**Sprint 1 (Planning):** Project requirement analysis, Git initialization, React setup, and tracing GeoJSON coordinates for Barishal routes.

Days 11-25	**Sprint 2 (Simulation):** Python FastAPI backend development, database schema design, and creating simulation loops for moving coordinates.
Days 26-40	**Sprint 3 (Real-Time):** Mapbox integration, 3D model rendering, and establishing Socket.IO WebSockets to stream data to the UI.
Days 41-55	**Sprint 4 (Intelligence):** Scikit-Learn integration for Self-Sharpening ETA, Geofencing state logic, and Red/Yellow capacity triggers.
Days 56-65	Admin Dashboard creation, automated SOS visual UI integration, rigorous system debugging, and WebSocket latency testing.
Days 66-70	Final cloud deployment, complete project documentation, presentation preparation, and final academic submission.

Conclusion

CampusFlow 3D is designed to address a critical operational challenge faced by the university campus: the reliance on static, blind transit dispatching. It brings a modern, proactive Digital Twin experience into the university environment.

Unlike a simple GPS tracker, CampusFlow 3D manages predictive logistics. For students, it provides a simple way to verify route availability, track intelligent ETAs, and avoid overloaded buses via visual capacity thresholds. For administrators, it offers a professional dashboard to monitor the entire fleet and gather historical data for route optimization.

Future enhancements may include scaling this exact software architecture to national commercial *poribohon* fleets, establishing a smart-transit grid for Bangladesh. In essence, CampusFlow 3D represents a highly feasible, deeply innovative vision: making transit easier to track, easier to manage, and more predictable for every student and administrator.